

Harikaran C

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EDUCATION

SSN College of Engineering

B.Tech. Information Technology, CGPA: 8.50 / 10

Chennai, India

Aug. 2023 – May 2027

Alwin International Public School (CBSE)

10th Grade: 474 / 500, 12th Grade: 482 / 500

JEE Mains (April 2023): 96 Percentile

Chennai, India

June 2019 – Aug. 2022

TECHNICAL SKILLS

Languages: Python, Java, C++ (Basics)

Frameworks/Tools: Flask, MySQL, MERN, REST API

Course Work: Data Structures and Algorithms, Competitive Programming, Object Oriented Programming, Computer Networks, Database Management, Operating Systems

Soft Skills: Problem-Solving, Adaptability, Collaboration, Time Management

PROJECTS

Ambulance Route Optimization System | *Flask, React, MongoDB, TomTom API* Jan. 2025 – Apr. 2025

- Built and Deployed a full-stack Flask-React web application for real-time ambulance routing with dynamic re-routing using TomTom Traffic & Routing APIs
- Developed an Optimal Hospital Finder to automatically select the nearest hospital with minimum ETA based on live traffic conditions
- Integrated MongoDB for data management, storing driver routes and histories to enable reliable tracking and analysis
- Designed an interactive React interface to visualize live navigation updates and enhance driver usability

Multimodal Sleep Stage Classification | *Python, TensorFlow/Keras, MNE, Scikit-learn* Sept. 2025 – Dec. 2025

- Developed an end-to-end deep learning pipeline for automated sleep stage classification from multimodal biosignals (EEG, EOG, EMG)
- Designed and trained a hybrid CNN-LSTM model to learn complex spatial-temporal features, achieving robust generalization across subjects
- Implemented advanced preprocessing using MNE for filtering, epoching, and normalization, enabling stable model convergence and high accuracy on the PhysioNet Sleep-EDF dataset

AlgoMart: Small Business | *Python, Flask, HTML/CSS, JavaScript, MySQL* Mar. 2024 – Jun. 2024

- Built a dynamic shopping website with secure user login, cart management, delivery estimation, and owner dashboard
- Used Binary Search Tree (BST) for optimized product search functionality
- Implemented PL/SQL triggers and procedures for automated order confirmation, inventory updates, and customer emails
- Enabled real-time inventory updates by the store owner through the owner dashboard

TECHNICAL ACHIEVEMENTS

Competitive Programming

Languages: Python, C++

- Codeforces** Max Rating: 1323, **Pupil**

LEADERSHIP

SSN Coding Club | *Deputy Lead, CP Vertical*

Aug. 2025 – Present

- Organized and mentored coding competitions, fostering peer learning and enhancing problem-solving culture within the club

SSN Photography Club | *Core Committee Member*

Jan. 2024 – Present

- Coordinated events and workshops while mentoring peers to enhance creative collaboration and technical skills

CERTIFICATIONS

- NPTEL Elite Silver Certification in Programming in Java** - IIT Kharagpur (12-week)
- NPTEL Elite Silver Certification in Getting Started with Competitive Programming** - IIT Gandhinagar (12-week)